Sprint 1 Report

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Sprint Objectives

The goal of Sprint 1 was to:

- Acquire and configure the Freenove 4WD Smart Car.
- Establish reliable communication with the Raspberry Pi.
- Verify baseline functionality of motors, sensors, and the camera.
- Confirm ability to upload and execute custom scripts.
- Begin mechanical redesign work (camera mount, ball mount, chassis redesign).

By the end of the sprint, we aimed to have a confirmed working hardware platform and initial groundwork for both software and mechanical development.

Steps to Achieve the Sprint Goal

1. *Hardware Setup & System Initialization*

- Received the pre-assembled Freenove 4WD Smart Car
- Powered the vehicle using the provided PCB and battery system
- Established SSH connection to the Raspberry Pi
- Configured required Freenove libraries
- Verified camera configuration and system interfaces

We successfully executed the provided Freenove demo programs, the motor test and the obstacle avoidance code. This confirmed that the drivetrain, ultrasonic sensor, camera module, and onboard PCB are functional.

Additionally, we verified that we could upload custom Python scripts to the Raspberry Pi and execute them successfully on the robot. Our custom control code enabled forward motion and responded to ultrasonic sensor input. The car comes to a hard stop when an obstacle is detected within the defined safety threshold and resumes forward motion once the path is clear.

2. *Software Development Progress*

- Developed a detailed draft structure of our intended autonomous control code
- Uploaded the draft code to the team GitHub repository

- Verified successful transfer and execution capability on the robot

At this stage:

- A basic forward-drive state was implemented
- Ultrasonic-based safety stopping logic was validated
- A simple state-machine structure was tested

During testing, the vehicle stops immediately when an obstacle is placed within range. Once the obstacle is removed and the measured distance exceeds the safety threshold, the system resumes forward motion.

It was observed that if the vehicle is on a high-friction surface, it requires a slight nudge to resume motion due to static friction. This is because the code does not yet include reversing logic. Also, ArUco-based navigation logic has been drafted but has not yet been integrated into the code.

3. Mechanical Redesign Development

As discussed and approved in our meeting with the project supervisor, the following design modifications were initiated:

Tank Drive Configuration

- Decision made to remove the rear two wheels.
- Transition to a 2-wheel differential tank-drive configuration.

Rear Ball Mount

- Designed a rear ball mount in SolidWorks.
- The ball mount includes two mounting holes to secure the chassis to the rear.
- Model is near completion.

Chassis Redesign

- Created a draft chassis model in SolidWorks.
- Front mounting will use the existing light sensor screw holes.
- Rear mounting will connect through the ball mount.
- Removing the 3-servo camera configuration to reduce wiring complexity and power consumption.

Camera Mount

- Began planning a fixed camera mount.
- Final design still in progress.

All mechanical components remain in the modelling stage and have not yet been printed.

Demo Link

A demo video link is provided below showing the Freenove obstacle avoidance program running successfully.

Link:
https://uwoca.sharepoint.com/:v:/s/MechaTron/IQA4jscMUhdzRp3xuJDSeo6zAfLW1X_ponBrOyUK7FQpYRI

Review & Retrospective

1. What Went Well During This Sprint?

Hardware Connectivity & Integration

We successfully established a connection to the vehicle. While initial communication with the Raspberry Pi was difficult, troubleshooting resulted in stable SSH access and verified system functionality.

System Functionality Verification

We confirmed that all hardware components are operational. Tests using the Freenove demo programs verified that the motors, sensors, camera, and PCB are functioning correctly.

Code Execution Control

We successfully implemented and tested basic autonomous safety behaviour using ultrasonic sensing. The vehicle drives forward, performs a hard stop when an obstacle is detected within a defined threshold, and resumes forward motion once clearance is restored.

Mechanical Redesign Progress

Mechanical design work began alongside software development. Preliminary SolidWorks models for the ball mount and chassis were created to address known design weaknesses.

2. What Didn't Go Well / What Can Be Improved?

Limited Autonomous Capability

While basic obstacle detection and stopping behaviour were achieved, the vehicle does not yet perform intelligent navigation. No reverse maneuver, rerouting logic, or ArUco-based target tracking is implemented.

Causes: Time constraints limited the ability to expand beyond basic forward/stop logic.

Improvements: Extend the control logic to include reverse recovery behaviour and integrate vision-based navigation.

Manufacturing Delays

Mechanical components were not manufactured during this period. The CAD designs remain in the draft phase and have not been 3D printed.

Improvements: Finalize all CAD models earlier in the next sprint and begin/finalize printing.

Team Coordination

We experienced challenges with task coordination. When technical blocks occurred, it was unclear if other members should assist or work independently. This led to some inefficiency between the software and mechanical tasks.

Improvements: Define clear short-term goals for each member.

Plan for the Next Sprint

- Start tasks earlier to recover time lost during previous delays.
- Use the upcoming break to focus on debugging and system integration.
- Finalize all CAD designs and begin the 3D printing process.
- Prepare a physical ArUco marker board for navigation testing.
- Begin integrating perception and motion control in small, manageable stages.